Interactive Image Mosaic Generator

Method

The Interactive Image Mosaic Generator reconstructs an input image using small tiles following a structured approach. First, the input image undergoes preprocessing, where it is resized to ensure its dimensions are proportional to the chosen tile size. Additionally, color quantization is applied using K-means clustering to reduce color variations, making it easier to match tiles.

Next, the image is divided into a grid based on the selected tile size (8×8, 16×16, or 32×32). Each grid cell is analyzed to determine its average color, which is then matched with the closest available tile using an optimized KDTree search. The best-matching tile is then resized and placed in the corresponding grid position, ensuring a seamless mosaic reconstruction.

A Gradio interface is implemented to provide user interaction, allowing users to upload an image and select the desired tile size. The processed mosaic image is then displayed alongside the original for comparison.

Performance Metric

To evaluate the quality of the mosaic reconstruction, two primary metrics are used. The Mean Squared Error (MSE) quantifies the pixel-wise difference between the original and mosaic images, with lower values indicating better similarity. Additionally, the Structural Similarity Index (SSIM) assesses visual similarity by measuring changes in structural information, where a higher score (closer to 1) signifies greater resemblance. These metrics collectively help in analyzing the effectiveness of the mosaic generation process.

Result

The mosaic image loses the contrast of the original image. I think the problem is caused by the color quantization and averaging color of tiles. The MSE is good, so the shape of the original image keeps well. However, the SSIM score is low and not higher than 0.05. The color of the mosaic image looks like blue and loses the contrast of the original image.